

Oblicz granicę jednostronną $\lim_{x \rightarrow -3^-} \frac{x+3}{\sqrt{x^2-9}}$

① podstawiamy -3 do wzoru

$$\lim_{x \rightarrow -3^-} \frac{x+3}{\sqrt{x^2-9}} = \left[\frac{0}{0} \right]$$

$$= \frac{\sqrt{(-3)^2-9}}{-3-3} = \frac{\sqrt{9-9}}{-6} = \frac{0}{-6} = 0$$

② usuwamy niewymierność

$$\lim_{x \rightarrow -3^-} \frac{x+3}{\sqrt{x^2-9}} \cdot \frac{\sqrt{x^2-9}}{\sqrt{x^2-9}} = \lim_{x \rightarrow -3^-} \frac{(x+3)(\sqrt{x^2-9})}{x^2-9} =$$

$$= \lim_{x \rightarrow -3^-} \frac{\cancel{(x+3)}(\sqrt{x^2-9})}{\cancel{(x+3)}(x-3)} = \lim_{x \rightarrow -3^-} \frac{\sqrt{x^2-9}}{x-3} =$$

odp: 0